

Average number of HI ionizations per primary particle

Intergalactic medium at $z = 10$, $x_e = 10^{-4}$

Computed from the `igm_losses` energy-loss prescription

Generated with an AI assistant, under the project Master Rules

Reproduced by `python3 yield_comparison.py`

Abstract

We compute the mean number of HI ionizations produced by one primary electron (10^2 – 10^{12} eV) or photon (10^1 – 10^5 eV) in the neutral IGM at $z = 10$ with a static ionized fraction $x_e = 10^{-4}$, counting every ionization anywhere in the cascade. **Electron propagation is computed with `igm_losses.py`:** seven competing loss mechanisms (adiabatic expansion, synchrotron, inverse Compton with the Klein–Nishina kernel, Coulomb, collisional excitation, collisional ionization, bremsstrahlung), with ionizations counted event by event against the RBED cross section of Kim et al. [1] and knock-on electrons followed explicitly through the BED singly-differential cross section of Kim & Rudd [2]. Discarding the energy inverse Compton gives to the CMB (**the loss route**) the yield saturates at $1.171 \times 10^{4\text{N_e_loss_sat}}$ ion pairs. Following the upscattered CMB photons that are reabsorbed (**the loss+IC route**) raises it to $4.911 \times 10^{6\text{N_e_loss_ic_sat}}$, a factor $419.5^{\text{lossic_over_loss_sat}}$. The deposition-fraction route used in earlier versions of this document (models A/B/C, built on the Shull & van Steenberg fit) is **obsolete**; it survives only as the independent cross-check that validated this one, agreeing to $1.071^{\text{lossic_over_depfit_1e12}}$ at saturation and to $1.017^{\text{loss_over_ic_min}}\text{--}1.102^{\text{loss_over_ic_max}}$ below 1 MeV.

1 Assumptions

1. **Target:** neutral hydrogen of the IGM at $z = 10$, number density $n_{\text{HI}} = 2.5287 \times 10^{-4} \text{ cm}^{-3\text{n_H_cm3_route2}}$, ionized fraction $x_e = 10^{-4\text{x_e_route2}}$ held static.
2. **Ionization threshold** $E_{\text{th}} = 13.5984 \text{ eV}^{\text{E_th_eV_route2}}$ (one Rydberg, as `igm_losses` defines it).
3. **Magnetic field** $B_0 = 1 \times 10^{-13} \text{ T}^{\text{B0_T_route2}}$ comoving (1 nG), giving $U_B/U_{\text{CMB}} \sim 10^{-7}$ at $z = 10$ — which is why synchrotron is negligible rather than assumed away.
4. **Expansion** at $H(z) = 4.4717 \times 10^{-17} \text{ s}^{-1\text{H_z_s_route2}}$ (Planck18, the cosmology `igm_losses` itself uses).
5. **Cascade:** knock-on electrons are followed explicitly. The secondary energy is drawn from the BED SDCS [2], not from a deposition fraction.
6. **One generation** of IC-secondary photons — derived, and exactly, in §2.1.
7. **Environmental parameters are adjustable** through `igm_config.py`; every value above is what the run actually used.

2 Method

An electron of kinetic energy K loses energy through seven channels; only the collisional-ionization channel makes ion pairs, and it makes them one at a time:

$$Y(K) = \int_{E_{\text{th}}}^K R(K') [1 + \langle Y(\varepsilon) \rangle_{p(\varepsilon|K')}] dK', \quad R(K') = \frac{n_{\text{HI}} v(K') \sigma_{\text{ion}}(K')}{L_{\text{total}}(K')} \quad (1)$$

R is the number of ionization events bought per eV given up, with every competing channel in the denominator; $p(\varepsilon|K')$ is the BED secondary spectrum. Equation (1) is **the loss route**. The energy handed to the CMB is discarded there; **the loss+IC route** follows it,

$$Y_E(K) = Y_D(K) + \int_0^K \frac{L_{\text{IC}}}{L_{\text{tot}}}(K') G(\gamma(K')) dK' \quad (2)$$

with G the ionizations per eV radiated, obtained by folding the Blumenthal & Gould [5] scattered-photon spectrum against the absorption probability and the photon yield. The IC branching ratio is the Klein–Nishina-corrected one from `igm_losses`; the photoelectron released by an absorbed secondary cascades with Y_D , not with a fit.

Why mixing a KN loss rate with a Thomson-limit photon spectrum is legitimate.

Checked, not assumed, at the energies that actually deliver the ionizations: at 5 % of the channel ($K = 2.0224 \times 10^7 \text{ eV}^{\text{Ewindow_lo_eV}}$) the KN factor is $1.0^{\text{FKN_window_lo}}$; at 95 % ($K = 3.5541 \times 10^8 \text{ eV}^{\text{Ewindow_hi_eV}}$) it is $0.9999^{\text{FKN_window_hi}}$. Corrections stay below 10^{-4} across the window.

2.1 Why one generation suffices

Every absorbed secondary photon has $\varepsilon_1 \lesssim 1218 \text{ eV}$, so its photoelectron has $\gamma = 1.0024$; reaching 13.6 eV from there would need a seed photon of $1310 kT$, Planck occupancy $\sim 10^{-569}$. On the tracked spectrum $G(\gamma) = 0$ identically.

3 The loss budget

mechanism	peak fraction of dK/dt	at K [eV]
inverse Compton (KN)	$1.000^{\text{peakfrac_compton}}$	$\geq 10^9$
collisional ionization	$0.7304^{\text{peakfrac_ionization}}$	$\sim 5 \times 10^3$
collisional excitation	$0.3532^{\text{peakfrac_excitation}}$	10^2
adiabatic expansion	$0.0929^{\text{peakfrac_adiabatic}}$	$3.046 \times 10^5^{\text{peakE_adiabatic_eV}}$
Coulomb	$0.002336^{\text{peakfrac_coulomb}}$	—
bremsstrahlung	$4.159 \times 10^{-5}^{\text{peakfrac_bremsstrahlung}}$	—
synchrotron	$1.176 \times 10^{-7}^{\text{peakfrac_synchrotron}}$	—

Adiabatic expansion reaches 9.2% and has no counterpart in the obsolete route, which had no expansion term at all. The last three channels stay below a quarter of a per cent everywhere — which is the quantitative reason the old two-channel treatment could ignore them and still land within 10 %.

4 Result

- W is *not* a constant: $40.37 \text{ eV}^{\text{W_loss_1e+02}}$ at 10^2 eV , $35.27 \text{ eV}^{\text{W_loss_1e+03}}$ at 10^3 eV , $45.40 \text{ eV}^{\text{W_loss_1e+05}}$ at 10^5 eV , where IC has begun to take the energy away.
- $E_{\text{crit}} = 1.4866 \times 10^5 \text{ eV}^{\text{E_crit_D_eV}}$, where Compton overtakes excitation plus ionization.
- Photon primaries: $N_\gamma = 3.077^{\text{N_gamma_loss_1e+02}}$ at 10^2 eV and $28.97^{\text{N_gamma_loss_1e+03}}$ at 10^3 eV — one ionization plus the photoelectron’s cascade, on the same prescription.

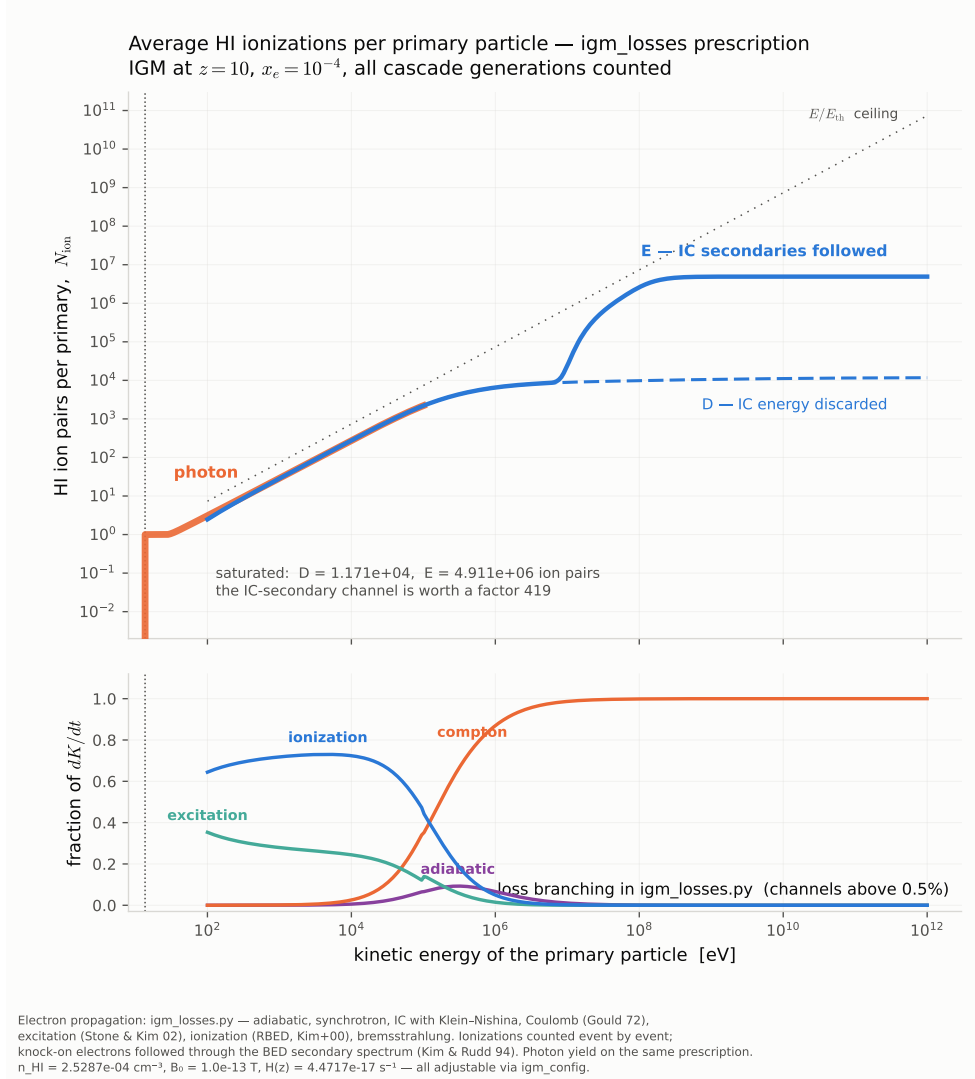


Figure 1: HI ion pairs per primary against primary kinetic energy, on the `igm_losses` prescription. *Upper*: the loss route (IC energy discarded) and the loss+IC route (IC secondaries followed), with the photon curve on the same prescription and the E/E_{th} ceiling. *Lower*: the seven-channel loss branching that produces the shape. Produced by `ionization_yield_igm.py`.

5 Validation against the obsolete route

The deposition-fraction route (models A/B/C; Shull & van Steenberg [7] via the Ricotti fit reproduced in Furlanetto & Stoeve [8]) shares nothing with this one but the physical scenario. It is retained for exactly one purpose:

- below 1 MeV, D/B lies between $1.017^{\text{loss_over_ic_min}}$ and $1.102^{\text{loss_over_ic_max}}$;
- at 10^{12} eV, $D/B = 1.32^{\text{loss_over_ic_1e12}}$, and the excess is Klein–Nishina: the obsolete route used the Thomson limit, where the KN factor is $0.8111^{\text{FKN_at_1e12}}$ at that energy;
- with the secondary channel on both sides, $E/C = 1.071^{\text{lossic_over_depfit_1e12}}$.

The two environments have since been *harmonised* on the route-1 values, at the user’s instruction: n_{H} now differs by $1.05 \times 10^{-10}^{\text{env_dn_H}}$, i.e. floating-point round-off, and the ionization threshold agrees exactly, $\Delta E_{\text{th}} = 0^{\text{env_dE_th}}$ — `igm_losses.py` was moved from one Rydberg to the NIST binding energy. Any residual disagreement between the routes is therefore method, not input. One input is *not* harmonised and is reported rather than hidden: $H(z)$, which the two routes

Table 1: N_{ion} per primary. The loss+IC route is the answer; D is shown to isolate the IC-secondary channel.

K [eV]	e^- (D)	e^- (E)	$W = K/N$ [eV]
10^2	$2.477^{N_{\text{e_loss_1e+02}}}$	$2.478^{N_{\text{e_loss_ic_1e+02}}}$	$40.37^{W_{\text{loss_1e+02}}}$
10^3	$28.35^{N_{\text{e_loss_1e+03}}}$	$28.36^{N_{\text{e_loss_ic_1e+03}}}$	$35.27^{W_{\text{loss_1e+03}}}$
10^4	—	—	$36.18^{W_{\text{loss_1e+04}}}$
10^5	—	—	$45.4^{W_{\text{loss_1e+05}}}$
10^6	$6523^{N_{\text{e_loss_1e+06}}}$	$6523^{N_{\text{e_loss_ic_1e+06}}}$	—
10^9	$1.058 \times 10^4^{N_{\text{e_loss_1e+09}}}$	$4.892 \times 10^6^{N_{\text{e_loss_ic_1e+09}}}$	—
10^{12}	$1.171 \times 10^4^{N_{\text{e_loss_1e+12}}}$	$4.911 \times 10^6^{N_{\text{e_loss_ic_1e+12}}}$	—

evaluate from the same Planck 18 cosmology by different code paths, differing by $0.001369^{\text{env_dH_z}}$. It reaches the yield only through the adiabatic loss term.

6 What was *not* checked

1. **No independent reimplementation** of `igm_losses` itself. The agreement above is with the obsolete route, which is a genuinely separate construction, but both were written in this project.
2. **Helium** is absent from the target in both routes.
3. x_e **is static**; no back-reaction of the cascade on the gas.
4. **Cosmological transport of the primary** is not modelled: “per primary” presupposes an interaction.
5. **A fixed- z snapshot.** `igm_losses` is built to integrate trajectories over cosmic time; here it is evaluated at $z = 10$ throughout. Following a real electron as it redshifts would change the adiabatic and IC terms.

Reproducibility. Deterministic; no random numbers. Run `python3 yield_comparison.py` for the registry and `python3 ionization_yield_igm.py` for the figure, then `python3 check_provenance.py ionization_yield.tex yield_comparison_results.json ionization_yield.log`.

References

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